



# basic education

Department:  
Basic Education  
**REPUBLIC OF SOUTH AFRICA**

**SENIOR CERTIFICATE EXAMINATIONS/  
NATIONAL SENIOR CERTIFICATE EXAMINATIONS  
*SENIORSERTIFIKAAT-EKSAMEN/  
NASIONALE SENIORSERTIFIKAAT-EKSAMEN***

**PHYSICAL SCIENCES: CHEMISTRY (P2)  
*FISIESE WETENSKAPPE: CHEMIE (V2)***

**MAY/JUNE 2025/*MEI/JUNIE 2025***

**MARKING GUIDELINES/*NASIENRIGLYNE***

**MARKS/*PUNTE*: 150**

**These marking guidelines consist of 22 pages.  
*Hierdie nasienriglyne bestaan uit 22 bladsye.***

**QUESTION 1/VRAAG 1**

- 1.1 B ✓✓ ACCEPT/AANVAAR 2 (2)
- 1.2 D ✓✓ (2)
- 1.3 D ✓✓ (2)
- 1.4 A ✓✓ (2)
- 1.5 B ✓✓ (2)
- 1.6 C ✓✓ (2)
- 1.7 D ✓✓ (2)
- 1.8 B ✓✓ (2)
- 1.9 C ✓✓ (2)
- 1.10 C ✓✓ (2)
- [20]**

**QUESTION 2/VRAAG 2**

- 2.1 Compounds with one or more multiple bonds between C atoms in the hydrocarbon chain. ✓✓ (2 or 0)  
*Verbindings met een of meer meervoudige bindings tussen C-atome in die koolwaterstofkettings. (2 of 0)*

**OR/OF**

A hydrocarbon with two or more bonds between the C-atoms.  
*'n Koolwaterstof met twee of meer bindings tussen die C-atome.*

**OR/OF**

Hydrocarbons containing not only single bonds between C atoms.  
*Koolwaterstowwe wat nie slegs enkelbindings tussen die C-atome bevat nie.*

**ACCEPT/AANVAAR:**

Compounds with one or more double/triple bonds between C atoms in the hydrocarbon chain.  
*Verbindings met een of meer dubbel/trippelbindings tussen C-atome in die koolwaterstofkettings.* (2)

- 2.2
- 2.2.1 E ✓ (1)
- 2.2.2 F ✓ (1)
- 2.3 Ketones/Ketone ✓  
Aldehydes/Aldehiede ✓ (2)

## 2.4 Tertiary/Tersiêre ✓

The hydroxyl group/functional group (-OH) is bonded to a C atom that is bonded to three other C atoms. ✓

*Die hidroksiel/funksionele groep (-OH) is gebind aan 'n C-atoom wat aan drie ander C-atome gebind is.*

**OR/OF**

The functional group ( $\begin{array}{c} | \\ -\text{C}- \\ | \\ \text{OH} \end{array}$ ) is bonded to three other C atoms.

*Die funksionele groep ( $\begin{array}{c} | \\ -\text{C}- \\ | \\ \text{OH} \end{array}$ ) is gebind aan drie ander C-atome.*

(2)

## 2.5

## 2.5.1

**Marking criteria:**

- Correct stem, i.e. hexane. ✓
- Both substituent (ethyl and iodo) correctly identified. ✓
- IUPAC name completely correct including numbering, sequence, hyphens and commas. ✓

**Nasienkriteria:**

- *Korrekte stam, d.i. heksaan.* ✓
- *Beide substituent (etiel en jodo) korrek geïdentifiseer.* ✓
- *IUPAC-naam heeltemal korrek insluitende nommering, volgorde, koppeltekens en kommas.* ✓

3-ethyl-4-iodohexane/3-etiel-4-jodoheksaan ✓✓✓

(3)

## 2.5.2

**Marking criteria/Nasienkriteria:**

- Correct stem and substituents: methyl and propanol ✓  
*Korrekte stam en substituent: metiel en propanol*
- IUPAC name completely correct including numbering, sequence, hyphens and commas. ✓  
*IUPAC-naam heeltemal korrek insluitende nommering, volgorde, koppeltekens en kommas.*

2-methylpropan-1-ol/ 2-methyl-1-propanol/ methylpropan-1-ol/  
methyl-1-propanol ✓✓  
*2-metielpropan-1-ol/ 2-metiel-1-propanol / metielpropan-1-ol/  
metiel-1-propanol*

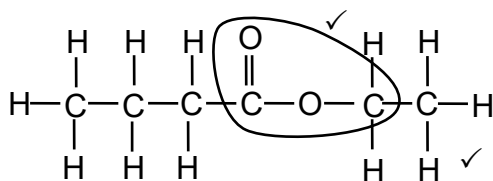
(2)

2.6

2.6.1 Esterification/Condensation/Verestering/Esterifikasie/Kondensasie ✓

(1)

2.6.2

**Marking criteria/Nasienkriteria:**

- Functional group correct. ✓  
*Funksionele groep korrek.*
- Whole structure correct. ✓  
*Hele struktuur korrek.*

**IF/INDIEN**

- More than one functional group/wrong functional group:  
*Meer as een funksionele groep/foutiewe funksionele groep:*  $0/2$
- If condensed structural formulae used/*Indien gekondenseerde struktuurformules gebruik:* Max./Maks.  $1/2$

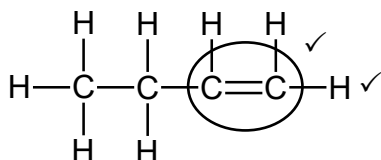
(2)

2.7

2.7.1  $C_2H_4O$  ✓

(1)

2.7.2

**Marking criteria/Nasienkriteria:**

- Correct functional group. ✓  
*Korrekte funksionele groep.*
- Whole structure correct. ✓  
*Hele struktuur korrek.*

**IF/INDIEN**

- More than one functional group/wrong functional group:  
*Meer as een funksionele groep/foutiewe funksionele groep:*  $0/2$
- If condensed structural formulae used/*Indien gekondenseerde struktuurformules gebruik:* Max./Maks.  $1/2$

(2)

**[19]**

**QUESTION 3/VRAAG 3**

- 3.1.1 (A series of organic) compounds that can be described by the same general formula. ✓✓ (2 or 0)

**OR**

(A series of organic) compounds in which one member differs from the next by a  $\text{CH}_2$  group.

(’n Reeks organiese) verbindings wat deur dieselfde algemene formule beskryf kan word. ✓✓ (2 of 0)

**OF**

(’n Reeks organiese) verbindings waarin die een lid van die volgende verskil met ’n  $\text{CH}_2$ -groep

(2)

3.1.2

- (a) Formyl (group)/Formiel(groep) ✓

(1)

(b)

**Marking criteria:**

- Correct chain length, i.e. Meth. ✓
- Everything else correct. ✓

**Nasienkriteria:**

- Korrekte kettinglengte d.i. Met. ✓
- Alles verder reg ✓

Methanal/Metanaal ✓✓

(2)

3.1.3

- (a) Homologous series/Functional group/Type of intermolecular forces/Straight chain/Atmospheric pressure ✓

Homoloë reeks/Funksionele groep/Tipe intermolekulêre kragte/ Reguitketting/ Atmosferiese druk

(1)

- (b) The boiling points of the carboxylic acids increase with an increase in the chain length/the number of carbon atoms/surface area/molecular mass./

**OR**

The boiling points of the carboxylic acids decrease with a decrease in the chain length/number of carbon atoms/surface area/molecular mass. ✓

Die kookpunte van die karboksielsure neem toe met ’n toename in die kettinglengte/aantal koolstofatome/reaksieoppervlak/molekulêre massa./

**OF**

Die kookpunte van die karboksielsure neem af met ’n afname in die kettinglengte/aantal koolstofatome/reaksieoppervlak/molekulêre massa.

(1)

(c)

**Marking criteria:**

For increasing or decreasing number of C atoms

- Compare the strength of intermolecular forces. ✓
- Compare the energy required to overcome intermolecular forces. ✓

**Nasienkriteria:***Vir toename of afname in aantal C-atome*

- *Vergelyk die sterkte van intermolekulêre kragte.* ✓
- *Vergelyk die energie benodig om intermolekulêre kragte te oorkom.* ✓

As the number of C atoms/chain length/surface area/contact area/molecular mass increases

- The strength of intermolecular/London/dispersion forces increases. ✓
- More energy is needed to overcome intermolecular forces/London/dispersion forces. ✓

**OR**

As the number of C atoms/chain length/surface area/contact area/molecular mass decreases

- The strength of intermolecular/London/dispersion forces decreases. ✓
- Less energy is needed to overcome intermolecular forces/London/dispersion forces. ✓

*Met toename in aantal C-atome/kettinglengte/reaksieoppervlak/kontakarea/molekulêre massa.*

- *Die sterkte van die die intermolekulêre kragte/Londonkragte/dispersiekragte neem toe.*
- *Meer energie word benodig om die intermolekulêre kragte/Londonkragte/dispersiekragte te oorkom/breek.*

**OF***Met afname in aantal C-atome/kettinglengte/reaksieoppervlak/kontakarea/molekulêre massa.*

- *Die sterkte van die die intermolekulêre kragte/Londonkragte/dispersiekragte neem af.*
- *Minder energie word benodig om die intermolekulêre kragte/Londonkragte/dispersiekragte te oorkom/breek*

(2)

3.1.4 75 °C ✓

(1)

3.2

**Marking criteria:**

- Higher than ✓
- State that carboxylic acids have more than one (two) site for hydrogen bonding and alcohols have one site for hydrogen bonding. ✓
- Comparing the strength of IMFs. ✓
- Comparing the number of molecules in a vapour phase at a given temperature/energy needed to overcome IMFs. ✓

**Nasienkriteria:**

- Hoër as ✓
- Stel dat karboksiesure het meer as een (twee) plekke vir waterstofbindings en dat alkohole een plek het vir waterstofbinding. ✓
- Vergelyk die sterkte van die IMK's/energie benodig om IMK's te oorkom. ✓
- Vergelyk die hoeveelheid molekules in die dampfase by 'n gegewe temperatuur/energie nodig om die IMK te oorkom. ✓

- Higher than ✓
- Compound B/CH<sub>3</sub>CH<sub>2</sub>CH<sub>2</sub>COOH/Carboxylic acid/Butanoic acid has (in addition to London forces and dipole-dipole forces), more than one site (two) for hydrogen bonding between molecules and compound A/CH<sub>3</sub>CH<sub>2</sub>CH<sub>2</sub>CH<sub>2</sub>CH<sub>2</sub>OH/Alcohol/Pentan-1-ol has (in addition to London forces and dipole-dipole forces) one site for hydrogen bonding between molecules. ✓
- Intermolecular forces in compound B/CH<sub>3</sub>CH<sub>2</sub>CH<sub>2</sub>COOH/Carboxylic acids/Butanoic acid are stronger. ✓
- More energy needed to overcome/break intermolecular forces in compound B/CH<sub>3</sub>CH<sub>2</sub>CH<sub>2</sub>COOH/Carboxylic acid/Butanoic acid.

**OR**

- At a given temperature there will be fewer molecules of compound B/CH<sub>3</sub>CH<sub>2</sub>CH<sub>2</sub>COOH/Carboxylic acids/Butanoic acid in the vapour phase. ✓

- Higher than ✓
- Compound A/CH<sub>3</sub>CH<sub>2</sub>CH<sub>2</sub>CH<sub>2</sub>CH<sub>2</sub>OH/Alcohol/Pentan-1-ol has (in addition to London forces and dipole-dipole forces) one site for hydrogen bonding between molecules and compound B/CH<sub>3</sub>CH<sub>2</sub>CH<sub>2</sub>COOH/Carboxylic acid/Butanoic acid has, (in addition to London forces and dipole-dipole forces), more than one site (two) for hydrogen bonding between molecules. ✓
- Intermolecular forces in compound A/CH<sub>3</sub>CH<sub>2</sub>CH<sub>2</sub>CH<sub>2</sub>CH<sub>2</sub>OH/Alcohol/Pentan-1-ol are weaker. ✓
- Less energy needed to overcome/break intermolecular forces in compound A/CH<sub>3</sub>CH<sub>2</sub>CH<sub>2</sub>CH<sub>2</sub>CH<sub>2</sub>OH/Pentan-1-ol/Alcohol.

**OR**

- At a given temperature there will be more molecules of compound A/CH<sub>3</sub>CH<sub>2</sub>CH<sub>2</sub>CH<sub>2</sub>CH<sub>2</sub>OH/Alcohol/ Pentan-1-ol in the vapour phase. ✓

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <ul style="list-style-type: none"> <li>• Hoër as</li> <li>• Verbinding <u>B/CH<sub>3</sub>CH<sub>2</sub>CH<sub>2</sub>COOH/Karboksielsure/Butanoësuur</u> het, (in toevoeging tot Londonkragte en dipool-dipoolkragte), <u>meer as een posisie</u> (twee) vir <u>waterstofbinding</u> tussen molekule en verbinding <u>A/CH<sub>3</sub>CH<sub>2</sub>CH<sub>2</sub>CH<sub>2</sub>CH<sub>2</sub>OH/Alkohol/Pentan-1-ol</u> het, (in toevoeging tot Londonkragte en dipool-dipoolkragte), <u>een posisie</u> vir <u>waterstofbinding</u> tussen molekule.</li> <li>• Intermolekulêre kragte in verbinding <u>B/CH<sub>3</sub>CH<sub>2</sub>CH<sub>2</sub>COOH/Karboksielsure/Butanoësuur</u> is sterker.</li> <li>• <u>Meer energie</u> word benodig om intermolekulêre kragte in verbinding <u>B/CH<sub>3</sub>CH<sub>2</sub>CH<sub>2</sub>COOH/Karboksielsure/Butanoësuur</u> te oorkom/breek</li> </ul> <p><b>OF</b></p> <ul style="list-style-type: none"> <li>• <u>By 'n gegewe temperatuur</u> sal daar <u>minder molekules</u> van verbinding <u>B/CH<sub>3</sub>CH<sub>2</sub>CH<sub>2</sub>COOH/Karboksielsure/Butanoësuur</u> in die <u>dampfase</u> wees.</li> </ul>                                                |  |
| <ul style="list-style-type: none"> <li>• Hoër as</li> <li>• Verbinding <u>A/CH<sub>3</sub>CH<sub>2</sub>CH<sub>2</sub>CH<sub>2</sub>CH<sub>2</sub>OH/Alkohol/Pentan-1-ol</u> het, (in toevoeging tot Londonkragte en dipool-dipoolkragte), <u>een posisie</u> vir <u>waterstofbinding</u> tussen molekule en verbinding <u>B/CH<sub>3</sub>CH<sub>2</sub>CH<sub>2</sub>COOH/Karboksielsure/Butanoësuur</u> het, (in toevoeging tot Londonkragte en dipool-dipoolkragte), <u>meer as een posisie</u> (twee) vir <u>waterstofbinding</u> tussen molekule.</li> <li>• Intermolekulêre kragte in verbinding <u>A/CH<sub>3</sub>CH<sub>2</sub>CH<sub>2</sub>CH<sub>2</sub>CH<sub>2</sub>OH/Alkohol/Pentan-1-ol</u> is swakker.</li> <li>• <u>Minder energie</u> word benodig om intermolekulêre kragte in <u>A/CH<sub>3</sub>CH<sub>2</sub>CH<sub>2</sub>CH<sub>2</sub>CH<sub>2</sub>OH/Alkohol/Pentan-1-ol</u> te oorkom/breek.</li> </ul> <p><b>OF</b></p> <ul style="list-style-type: none"> <li>• <u>By 'n gegewe temperatuur</u> sal daar <u>meer molekules</u> van verbinding <u>A/CH<sub>3</sub>CH<sub>2</sub>CH<sub>2</sub>CH<sub>2</sub>CH<sub>2</sub>OH/Alkohol/Pentan-1-ol</u> in die <u>dampfase</u> wees.</li> </ul> |  |

(4)  
[14]

**QUESTION 4/VRAAG 4**

4.1

4.1.1 Hydrogenation/*Hidrogenering/Hidrogenasie* ✓ (1)4.1.2 Dehydration/*Dehidrasie/Dehidratering* ✓ (1)

4.2

**Marking criteria:**

- Correct chain length, i.e. But. ✓
- Everything else correct: IUPAC name completely correct including numbering. ✓

**Nasienkriteria:**

- *Korrekte kettinglengte d.i. But.* ✓
- *Alles verder reg: IUPAC-naam heeltemal korrek insluitende nommering.* ✓

Butan-1-ol/1-butanol ✓✓

(2)

4.3

4.3.1

**Marking criteria/Nasienkriteria:**

- Whole condensed structural formula of alkene correct. ✓  
*Hele gekondenseerde struktuurformule van die alkeen korrek.*
- H<sub>2</sub>O. ✓
- Hydroxyl group/OH. ✓  
*Hidroksielgroep/OH.*
- Whole condensed structural formula of alcohol correct (OH on second C-atom). ✓  
*Hele gekondenseerde struktuurformule van alkohol korrek (OH op tweede C-atoom)*

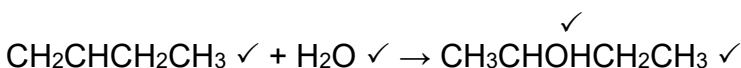
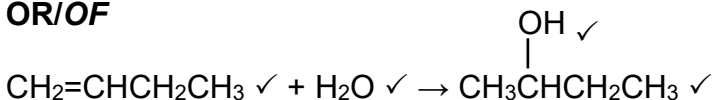
**IF/INDIEN**

- Any additional reactants or products /*Enige addisionele reaktanse of produkte:*  
Deduct 1 mark/*Trek 1 punt af.*

- Structural formulae used/*Struktuurformule gebruik.* Max./Maks.  $\frac{3}{4}$

- Molecular formulae used/*Molekulêre formule gebruik.* Max./Maks.  $\frac{1}{4}$

- Only reactants without arrow/*Slegs reaktanse sonder pyl* Max/Maks.  $\frac{2}{4}$

Marking rule 6.3.10/*Nasienreël 6.3.10***OR/OF**

(4)

4.3.2 Sulphuric acid/H<sub>2</sub>SO<sub>4</sub>/Phosphoric acid/H<sub>3</sub>PO<sub>4</sub>/Swawelsuur/Fosforsuur ✓

(1)

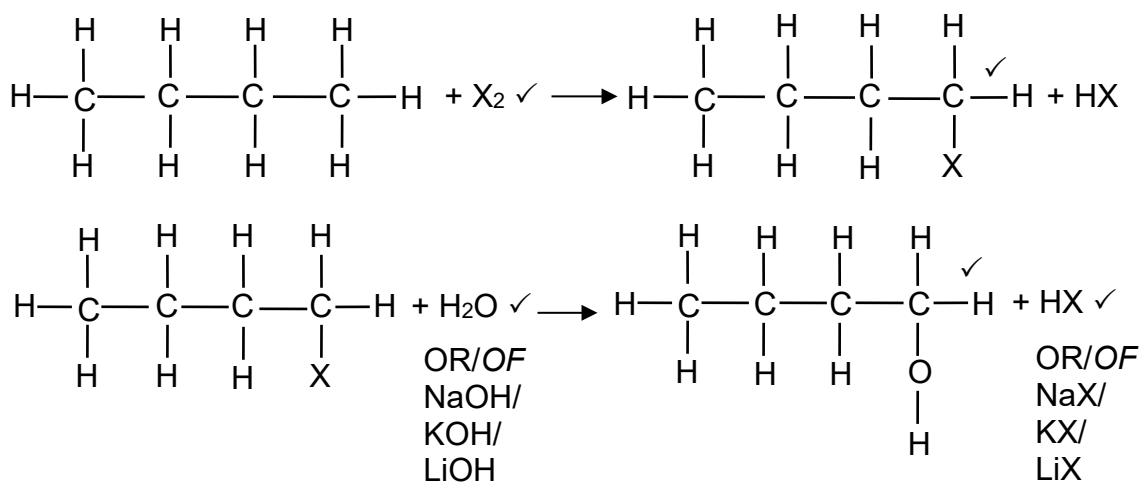
4.4

**Marking criteria/Nasienkriteria:**

- $X_2 = Br_2/Cl_2$ . ✓
- Whole structural formula of haloalkane correct. ✓  
*Hele struktuurformule van haloalkaan korrek.*
- $H_2O/NaOH/KOH/LiOH$ . ✓
- Whole structural formula of alcohol correct. ✓  
*Hele struktuurformule van alkohol korrek.*
- $HX/NaX/KX/LiX$  where/waar  $X = Br/Cl$  ✓

**IF/INDIEN**

- Any additional reactants or products /Enige addisionele reaktanse of produkte:  
Max./Maks.  $\frac{4}{5}$
- Condensed structural formulae used/Gekondenseerde struktuurformule gebruik:  
deduct 1 mark/trek 1 punt af.
- If inorganic product does not correspond with inorganic reactant: no mark for inorganic product./Indien anorganiese produk nie met die anorganiese reaktans ooreenstem nie, geen punt vir anorganiese produk.
- Molecular formulae used:/Molekulêre formule gebruik: Max./Maks.  $\frac{3}{5}$
- Marking rule 6.3.10/Nasienreël 6.3.10



(5)

4.5



Ignore phases./Ignoreer fases.

**Marking criteria/Nasienkriteria:**

- $C_4H_{10}$  ✓  $O_2$ ,  $CO_2$  and/en  $H_2O$  ✓ Balancing/Balansering ✓
- Ignore double arrows./Ignoreer dubbelpyle.
- Marking rule 6.3.10/Nasienreël 6.3.10.

**IF/INDIEN:**

- Structural formulae  $C_4H_{10}$  used:/Struktuurformule  $C_4H_{10}$  gebruik: Max./Maks.  $\frac{2}{3}$
- Balancing mark only if everything else is correct/  
*Balanseringspunt slegs indien alles korrek.*

(3)

[17]

**QUESTION 5/VRAAG 5**

5.1

**NOTE/NOTA**Give the mark for per unit time only if in context of reaction rate.*Gee die punt vir per eenheidtyd slegs indien in konteks van reaksietempo.***ANY ONE:**

- Change in concentration ✓ of products/reactants per (unit) time. ✓
- Change in amount/number of moles/volume/mass of products or reactants per (unit) time.
- Amount/number of moles/volume/mass of products formed/reactants used per (unit) time.
- Rate of change in concentration/amount/number of moles/volume/mass. ✓✓ **(2 or 0)**

**ENIGE EEN:**

- Verandering in konsentrasie ✓ van produkte/reaktanse per (eenheid)tyd. ✓
- Verandering in hoeveelheid/getal mol/volume/massa van produkte of reaktanse per (eenheid)tyd.
- Hoeveelheid/getal mol/volume/massa van produkte gevorm/reaktanse gebruik per (eenheid)tyd.
- Tempo van verandering in konsentrasie/ hoeveelheid/getal mol/ volume/ massa. ✓✓ **(2 of 0)**

(2)

5.2

**ANY ONE:**

Temperature ✓/

(Initial) amount/Mass of magnesium carbonate/Surface area

**ENIGE EEN:**

Temperatuur ✓/

(Aanvanklike) hoeveelheid/Massa van magnesiumkarbonaat/

Reaksieoppervlak

(1)

5.3

CO<sub>2</sub>/gas escapes from the reaction flask. ✓CO<sub>2</sub> /gas ontsnap uit die reaksiefles. ✓

(1)

5.4

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Marking criteria:</b></p> <p>(a) Mass subtraction ✓</p> <p>(b) Formula: <math>n = \frac{m}{M}</math> or <math>V = nV_m</math> ✓</p> <p>(c) Substitute <math>M = 44 \text{ g} \cdot \text{mol}^{-1}</math> in<br/> <math>n(\text{CO}_2) = \frac{m}{M}</math> with <math>m(\text{CO}_2)</math> from (a) ✓</p> <p>(d) Substitute <math>24,5 \text{ dm}^3</math> in <math>V = nV_m</math> with <math>n(\text{CO}_2)</math> ✓</p> <p>(e) Substitute <math>V_{\text{CO}_2}</math> and 120 in rate formula ✓</p> <p>(f) Final correct answer:<br/> <math>2,92 \times 10^{-3} (\text{dm}^3 \cdot \text{s}^{-1})</math> ✓<br/> Range: <math>2,08 \times 10^{-3}</math> to <math>3 \times 10^{-3}</math></p> | <p><math>m(\text{CO}_2) = 144,5 - 143,87 \checkmark \text{ (a)}</math><br/> <math>= 0,63 \text{ g}</math></p> <p><math>n(\text{CO}_2) = \frac{m}{M} \checkmark \text{ (b)}</math><br/> <math>= \frac{0,63}{44} \checkmark \text{ (c)}</math><br/> <math>= 1,43 \times 10^{-2} \text{ mol}</math></p> <p><math>V(\text{CO}_2) = nV_m</math><br/> <math>= (1,43 \times 10^{-2})(24,5) \checkmark \text{ (d)}</math><br/> <math>= 0,35 \text{ dm}^3</math></p> <p>Ave rate/<br/> gem tempo <math>= \frac{\Delta V(\text{CO}_2)}{\Delta t}</math><br/> <math>= \frac{0,35 - (0)}{120 - (0)} \checkmark \text{ (e)}</math><br/> <math>= 2,92 \times 10^{-3} (\text{dm}^3 \cdot \text{s}^{-1}) \checkmark \text{ (f)}</math></p> |
| <p><b>Nasienkriteria:</b></p> <p>(a) Aftek van massas. ✓</p> <p>(b) Formula: <math>n = \frac{m}{M}</math> or <math>V = nV_m</math> ✓</p> <p>(c) Vervang <math>M = 44 \text{ g} \cdot \text{mol}^{-1}</math> in<br/> <math>n(\text{CO}_2) = \frac{m}{M}</math> met <math>m(\text{CO}_2)</math> van (a) ✓</p> <p>(d) Vervang <math>24,5 \text{ dm}^3</math> in <math>V = nV_m</math> met <math>n(\text{CO}_2)</math> ✓</p> <p>(e) Vervang <math>V_{\text{CO}_2}</math> en 120 in tempoformule ✓</p> <p>(f) Finale korrekte antwoord:<br/> <math>2,92 \times 10^{-3} (\text{dm}^3 \cdot \text{s}^{-1})</math> ✓<br/> Gebied: <math>2,08 \times 10^{-3}</math> tot <math>3 \times 10^{-3}</math></p>         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

(6)

5.5

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Marking criteria:</b></p> <ul style="list-style-type: none"> <li>• A ✓</li> <li>• Comparison of the curves of the graph ✓</li> <li>• Comparison of concentration of <math>\text{HCl}</math> (from table) ✓</li> <li>• Explanation of collision theory for LOWER concentration ✓✓</li> </ul> <p><b>Nasienkriteria:</b></p> <ul style="list-style-type: none"> <li>• A ✓</li> <li>• Vergelyk die kurwes van grafiek ✓</li> <li>• Vergelyk die konsentrasie van <math>\text{HCl}</math> (vanaf tabel) ✓</li> <li>• Verduidelik botsingsteorie vir LAER konsentrasie ✓✓</li> </ul> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

A ✓

- Gradient is least steep/lowest reaction rate/least amount of gas produced in 120 s. ✓
- Lowest concentration of  $\text{HCl}(\text{aq})$ . ✓
- Least/Less particles per unit volume. ✓
- Least/Less effective collisions per unit time/second. ✓ **OR**  
Lowest/Lower frequency of effective collisions.
- *Gradient is die laagste/laagste reaksietempo/minste hoeveelheid gas geproduseer in 120 s.* ✓
- Laagste konsentrasie van  $\text{HCl}(\text{aq})$ . ✓
- Minste/Minder deeltjies per eenheidsvolume. ✓
- Minste/Minder effektiewe botsings per eenheidstyd/sekonde. ✓ **OF**  
Laagste/Laer frekwensie van effektiewe botsings.

(5)

5.6

The same/ *Dieselfde* ✓The same amount  $\text{MgCO}_3$  is used in each experiment. ✓*Dieselfde hoeveelheid  $\text{MgCO}_3$  is gebruik in elke eksperiment.*

(2)

[17]

**QUESTION 6/VRAAG 6**

6.1

6.1.1 Remains the same/*Bly dieselfde* ✓ (1)6.1.2 Decreases/*Neem af* ✓ (1)6.1.3 Remains the same/*Bly dieselfde* ✓ (1)

- 6.2
- Decrease in pressure favours the reaction that produces a greater number of moles/amount of gas. ✓  
*'n Verlagings in druk bevoordeel die reaksie wat 'n groter aantal mol/hoeveelheid gas produseer.*
  - Forward reaction is favoured. ✓ / [CO] increases AND [CO<sub>2</sub>] decreases  
*Voorwaartse reaksie word bevoordeel. / [CO] neem toe EN [CO<sub>2</sub>] neem af* (2)

6.3

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Marking criteria:</b></p> <p>(a) Substitute 44 in <math>n = \frac{m}{M}</math> ✓</p> <p>(b) Change in mass of carbon:<br/><math>m(C_i) - m(C_f)/n(C_i) - n(C_f)</math> ✓</p> <p>(c) Substitute 12 in <math>n = \frac{m}{M}</math> ✓</p> <p>(d) Use mole ratio 1:1 ✓</p> <p>(e) <math>n(\text{CO}_2)_{\text{eq}} = n(\text{CO}_2)_{\text{initial}} - n(\text{CO}_2)_{\text{used}}</math> OR<br/><math>m(\text{CO}_2)_{\text{eq}} = m(\text{CO}_2)_{\text{initial}} - m(\text{CO}_2)_{\text{used}}</math> ✓</p> <p>(f) Final answer: 6,16 g ✓<br/>RANGE: 6 to 6,16 g</p> <p><b>NOTE:</b><br/>If (b) <math>\Delta m(C)</math> or <math>\Delta n(C)</math> is not calculated<br/>max 2/6</p>                                                                                                                                                                                    | <p><b>Nasienkriteria:</b></p> <p>(a) Vervang 44 in <math>n = \frac{m}{M}</math> ✓</p> <p>(b) Verandering in massa:<br/><math>m(C_i) - m(C_f)/n(C_i) - n(C_f)</math> ✓</p> <p>(c) Vervang 12 in <math>n = \frac{m}{M}</math> ✓</p> <p>(d) Gebruik molverhouding 1:1 ✓</p> <p>(e) <math>n(\text{CO}_2)_{\text{ewe}} = n(\text{CO}_2)_{\text{begin}} - n(\text{CO}_2)_{\text{gebruik}}</math> OF<br/><math>m(\text{CO}_2)_{\text{ewe}} = m(\text{CO}_2)_{\text{begin}} - m(\text{CO}_2)_{\text{gebruik}}</math> ✓</p> <p>(f) Finale antwoord: 6,16 g ✓<br/>GEBIED: 6 tot 6,16 g</p> <p><b>NOTA:</b><br/>Indien (b) <math>\Delta m(C)</math> of <math>\Delta n(C)</math> nie bereken<br/>maks 2/6</p>                                            |
| <p><b>OPTION 1/OPSIE 1:</b></p> <p><math>\Delta m(C) = 14 - 4,44</math> ✓ (b)<br/><math>= 9,56 \text{ g}</math></p> <p><math>n(\text{CO}_2)_{\text{initially}} = \frac{m}{M}</math><br/><math>= \frac{41,2}{44}</math> ✓ (a)<br/><math>= 0,94 \text{ mol (0,936)}</math></p> <p><math>n(C)_{\text{used}} = \frac{m}{M}</math><br/><math>= \frac{9,56}{12}</math> ✓ (c)<br/><math>= 0,80 \text{ mol (0,797)}</math></p> <p><math>n(\text{CO}_2)_{\text{used}} = n(C)</math><br/><math>= 0,80 \text{ mol (0,797)}</math> ✓ (d)</p> <p><math>n(\text{CO}_2)_{\text{eq}} = n(\text{CO}_2)_{\text{initially}} - n(\text{CO}_2)_{\text{used}}</math><br/><math>= 0,94 - 0,80</math> ✓ (e)<br/><math>= 0,14 \text{ mol}</math></p> <p><math>n(\text{CO}_2) = \frac{m}{M}</math><br/><math>0,14 = \frac{m}{44}</math></p> <p><math>X = m(\text{CO}_2) = 6,16 \text{ (g)}</math> ✓ (f)</p> | <p><b>OPTION 2/OPSIE 2:</b></p> <p><math>\Delta m(C) = 14 - 4,44</math> ✓ (b)<br/><math>= 9,56 \text{ g}</math></p> <p><math>n(C)_{\text{used}} = \frac{m}{M}</math><br/><math>n(C)_{\text{used}} = \frac{9,56}{12}</math> ✓ (c)<br/><math>= 0,80 \text{ mol (0,797)}</math></p> <p><math>n(\text{CO}_2)_{\text{used}} = n(C)</math><br/><math>= 0,80 \text{ mol (0,797)}</math> ✓ (d)</p> <p><math>n(\text{CO}_2) = \frac{m}{M}</math><br/><math>0,80 = \frac{m}{44}</math> ✓ (a)<br/><math>m(\text{CO}_2) = 35,05 \text{ g}</math></p> <p><math>m(\text{CO}_2)_{\text{eq}} = m(\text{CO}_2)_{\text{initially}} - m(\text{CO}_2)_{\text{used}}</math><br/><math>= 41,2 - 35,05</math> ✓ (e)<br/><math>X = 6,15 \text{ (g)}</math> ✓ (f)</p> |

**OPTION 3/OPSIE 3:**

$$\begin{aligned}
 n(\text{CO}_2)_{\text{initially}} &= \frac{m}{M} \\
 &= \frac{41,2}{44} \checkmark \text{ (a)} \\
 &= 0,94 \text{ mol (0,936)}
 \end{aligned}$$

$$\begin{aligned}
 n(\text{C})_{\text{used}} &= \frac{9,56}{12} \checkmark \text{ (c)} \\
 &= 0,80 \text{ mol (0,797)}
 \end{aligned}$$

|                                                                      | C                     | CO <sub>2</sub>       |
|----------------------------------------------------------------------|-----------------------|-----------------------|
| Ratio/ <i>Verhouding</i>                                             | 1                     | 1                     |
| Initial quantity (mol)<br><i>Aanvangshoeveelheid (mol)</i>           | 1,17                  | 0,936                 |
| Change (mol)<br><i>Verandering (mol)</i>                             | 0,8                   | 0,8 $\checkmark$ (d)  |
| Quantity at equilibrium (mol)/<br><i>Hoeveelheid by ewewig (mol)</i> | 0,37 $\checkmark$ (b) | 0,14 $\checkmark$ (e) |

$$n(\text{CO}_2) = 0,139 \text{ mol}$$

$$m(\text{CO}_2) = 0,139 (44)$$

$$X = 6,16 \text{ (g)} \checkmark \text{ (f)}$$

(6)

6.4

**POSITIVE MARKING FROM QUESTION 6.3:****POSITIEWE NASIEN VANAF VRAAG 6.3:****Marking criteria**

- (a) Use of ratio  $n(\text{CO}_2) : n(\text{CO}) = 1:2$ . ✓  
 (b) Divide by  $3 \text{ dm}^3$ . ✓  
 (c) Correct  $K_c$  expression (formulae in square brackets). ✓  
 (d) Substitute of concentration into  $K_c$  expression. ✓  
 (e) Final answer: 5,98 ✓  
 RANGE: 5,98 – 7,29

**Nasienkriteria:**

- (a) Gebruik verhouding  $n(\text{CO}_2) : n(\text{CO}) = 1:2$ . ✓  
 (b) Deel deur  $3 \text{ dm}^3$ . ✓  
 (c) Korrekte  $K_c$  uitdrukking (formules in vierkantige hakies). ✓  
 (d) Vervang konsentrasies in korrekte  $K_c$  uitdrukking. ✓  
 (e) Finale antwoord: 5,98 ✓  
 GEBIED: 5,98 – 7,29

**NOTE/NOTA:**

Mark calculations of this question that may be done in QUESTION 6.3.

Merk berekeninge van hierdie vraag wat in VRAAG 6.3 gedoen is.

**CALCULATIONS USING NUMBER OF MOLES****BEREKENINGE WAT AANTAL MOL GEBRUIK****OPTION 1/OPSIE 1:**

$$n(\text{CO}_2)_{\text{initial}} = \frac{m}{M}$$

$$= \frac{41,2}{44}$$

$$= 0,936 \text{ mol}$$

|                                                                                                                              | $\text{CO}_2$ | $\text{CO}$ |
|------------------------------------------------------------------------------------------------------------------------------|---------------|-------------|
| Ratio/Verhouding                                                                                                             | 1             | 2           |
| Initial quantity (mol)<br>Aanvangshoeveelheid (mol)                                                                          | 0,936         | 0           |
| Change (mol)<br>Verandering (mol)                                                                                            | 0,8           | 1,6         |
| Quantity at equilibrium (mol)/<br>Hoeveelheid by ewewig (mol)                                                                | 0,14          | 1,6         |
| Equilibrium concentration ( $\text{mol} \cdot \text{dm}^{-3}$ )<br>Ewewigskonsentrasie ( $\text{mol} \cdot \text{dm}^{-3}$ ) | 0,047         | 0,53        |

✓ (a)

 Divide by/deel  
 deur 3 ✓ (b)

$$K_c = \frac{[\text{CO}]^2}{[\text{CO}_2]}$$

$$= \frac{(0,53)^2}{0,047}$$

$$= 5,98 \text{ ✓ (e)}$$

Wrong  $K_c$  expressionVerkeerde  $K_c$ -uitdrukking: Max./Maks.  $\frac{2}{5}$ No  $K_c$  expression/Geen  $K_c$ - uitdrukking:  $\frac{4}{5}$

**CALCULATIONS USING CONCENTRATION**  
**BEREKENINGE WAT KONSENTRASIE GEBRUIK**

**OPTION 2/OPSIE 2:**

$$c(\text{CO}_2) = \frac{m}{MV}$$

$$= \frac{41,2}{(44)(3)}$$

$$= 0,31 \text{ mol} \cdot \text{dm}^{-3}$$

Divide by 3 ✓ (b)

$$c = \frac{n}{V}$$

$$= \frac{0,8}{3}$$

$$= 0,27 \text{ mol} \cdot \text{dm}^{-3} (0,267)$$

|                                                                                                        | CO <sub>2</sub> | CO   |
|--------------------------------------------------------------------------------------------------------|-----------------|------|
| Ratio/Verhouding                                                                                       | 1               | 2    |
| Initial concentration (mol·dm <sup>-3</sup> )<br>Aanvangskonsentrasie (mol·dm <sup>-3</sup> )          | 0,31            | 0    |
| Change in concentration (mol·dm <sup>-3</sup> )<br>Verandering in konsentrasie (mol·dm <sup>-3</sup> ) | 0,27            | 0,54 |
| Equilibrium concentration (mol·dm <sup>-3</sup> )<br>Ewewigskonsentrasie (mol·dm <sup>-3</sup> )       | 0,04            | 0,54 |

✓ (a)

$$K_c = \frac{[\text{CO}]^2}{[\text{CO}_2]} \quad \checkmark \text{ (c)}$$

$$= \frac{(0,54)^2}{0,04} \quad \checkmark \text{ (d)}$$

$$= 7,29 \quad \checkmark \text{ (e)}$$

Wrong K<sub>c</sub> expression

Verkeerde K<sub>c</sub>-uitdrukking: Max./Maks. 2/5

No K<sub>c</sub> expression/Geen K<sub>c</sub>-uitdrukking: 4/5

**OPTION 3/OPSIE 3:**

$$n(\text{CO}_2)_{\text{initial}} = \frac{m}{M}$$

$$= \frac{41,2}{44}$$

$$= 0,936 \text{ mol}$$

$$\Delta n(\text{CO}_2) = 0,8 \text{ mol}$$

$$n(\text{CO}_2)_{\text{eqm}} = n(\text{CO}_2)_{\text{initial}} - \Delta n(\text{CO}_2)$$

$$= 0,936 - 0,8$$

$$= 0,136 \text{ mol}$$

$$n(\text{CO})_{\text{formed}} = 2\Delta n(\text{CO}_2)_{\text{used}} \checkmark \text{ (a)} = 1,6 \text{ mol}$$

$$n(\text{CO})_{\text{eqm}} = \Delta n(\text{CO})_{\text{formed}} = 1,6 \text{ mol}$$

$$\left. \begin{aligned} [\text{CO}_2]_{\text{eqm}} &= \frac{0,136}{3} = 4,53 \times 10^{-2} \text{ mol} \cdot \text{dm}^{-3} \\ [\text{CO}]_{\text{eqm}} &= \frac{1,6}{3} = 0,53 \text{ mol} \cdot \text{dm}^{-3} \end{aligned} \right\} \checkmark \text{ (b)}$$

$$K_c = \frac{[\text{CO}]^2}{[\text{CO}_2]} \checkmark \text{ (c)}$$

$$= \frac{(0,53)^2}{4,53 \times 10^{-2}} \checkmark \text{ (d)}$$

$$= 6,2 \checkmark \text{ (e)}$$

Wrong  $K_c$  expression

Verkeerde  $K_c$ -uitdrukking: Max./Maks.  $\frac{2}{5}$

No  $K_c$  expression/Geen  $K_c$ - uitdrukking:  $\frac{4}{5}$

(5)

6.5 Y ✓✓

(2)

6.6 Remains the same/Bly dieselfde ✓

(1)

**[19]**

**QUESTION 7/VRAAG 7**

7.1 An acid produces hydrogen ions /H<sup>+</sup>/hydronium ions/ H<sub>3</sub>O<sup>+</sup> in aqueous solution/water. ✓✓ (2 or 0)  
 'n Suur is 'n stof wat waterstofione/H<sup>+</sup>/hidroniumione/H<sub>3</sub>O<sup>+</sup> vorm in waterige oplossing/water. (2 of 0) (2)

7.2  
 7.2.1 (COOH)<sub>2</sub> ✓ (1)

7.2.2 NaCl ✓ (1)

7.2.3 HCO<sub>3</sub><sup>-</sup> ✓  
**OR/OF** NH<sub>3</sub> (1)

7.2.4 NaOH ✓✓  
**OR/OF** Mg(OH)<sub>2</sub> (2)

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7.3 | <p><b>Marking criteria:</b><br/> <b>(a)</b> Calculate n(H<sub>x</sub>Y) ✓<br/> <b>(b)</b> Calculate n(NaOH) ✓<br/> <b>(c)</b> Final answer: x = 2 ✓<br/> <b>(d)</b> Reactants ✓ Products ✓ Balancing ✓<br/> <b>NOTE:</b> Ignore ⇌ and phases<br/>         Marking rule 6.3.10</p>                                                                                                                                                                                                                                                                                                              | <p><b>Nasienkriteria:</b><br/> <b>(a)</b> Bereken n(H<sub>x</sub>Y) ✓<br/> <b>(b)</b> Bereken n(NaOH) ✓<br/> <b>(c)</b> Finale antwoord: x = 2 ✓<br/> <b>(d)</b> Reaktanse ✓ Produkte ✓ Balansering ✓<br/> <b>NOTA:</b> Ignoreer ⇌ en fases<br/>         Nasienreël 6.3.10</p>                                                                                                                                                                                                                                                                                                                             |
|     | <p><b>OPTION 1/OPSIE 1:</b><br/> <math>n = cV</math><br/> <math>n_{\text{acid}} = (0,11)(0,02364) \checkmark \text{ (a)}</math><br/> <math>= 2,6 \times 10^{-3}</math><br/> <math>n_{\text{base}} = (0,26)(0,02) \checkmark \text{ (b)}</math><br/> <math>= 5,2 \times 10^{-3} \quad (0,0052)</math><br/> <math display="block">\frac{n(\text{H}_x\text{Y})}{n(\text{NaOH})} = \frac{n_a}{n_b}</math> <math display="block">\frac{2,6 \times 10^{-3}}{5,2 \times 10^{-3}} = \frac{1}{n_b}</math> <math display="block">n_b = 2</math> <math>\therefore x = 2 \checkmark \text{ (c)}</math></p> | <p><b>OPTION 2/OPSIE 2:</b><br/> <math display="block">\frac{c_a V_a}{c_b V_b} = \frac{n_a}{n_b}</math> <math display="block">\text{(a)} \checkmark \frac{(23,64)(0,11)}{(20)(0,26)} = \frac{1}{n_b}</math> <math display="block">\text{(b)} \checkmark \frac{(20)(0,26)}{n_b} = 1</math> <math display="block">n_b = 2</math> <math>\therefore x = 2 \checkmark \text{ (c)}</math></p> <div style="border: 1px solid black; padding: 10px; margin: 10px auto; width: fit-content;"> <p>H<sub>x</sub>Y : NaOH<br/> <math>2,6 \times 10^{-3} : 5,2 \times 10^{-3}</math><br/> <math>1 : 2</math></p> </div> |
|     | <p><u>H<sub>2</sub>Y(aq) + 2NaOH(aq) ✓ → Na<sub>2</sub>Y (aq) + 2H<sub>2</sub>O(l) ✓</u> Bal ✓ <b>(d)</b> (6)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

7.4

**Marking criteria:**

- (a) Any formula:  $\text{pH} = -\log[\text{H}_3\text{O}^+]/\text{pH} = -\log[\text{H}^+]/[\text{H}_3\text{O}^+] = 10^{-\text{pH}}$  ✓  
 (b) Substitute 1,61 in  $\text{pH} = -\log[\text{H}_3\text{O}^+]$  ✓  
 (c) Calculate  $n(\text{HCl})_{\text{unused}}$  using  $c = \frac{n}{V}$  ✓  
 (d) Calculate  $n(\text{HCl})_{\text{initial}}$  using  $c = \frac{n}{V}$  ✓  
 (e) Calculate  $n(\text{HCl})_{\text{used}} = n(\text{HCl})_{\text{initial}} - n(\text{HCl})_{\text{unused}}$  ✓  
 (f) Using ratio 1:2 with **USED** HCl from of (e) to calculate  $n(\text{CaCO}_3)$  ✓  
 (g) Substitute 100 AND  $n(\text{CaCO}_3)$  from (f) in  $n = \frac{m}{M}$  ✓  
 (h) Mass of impurity =  $m_{\text{sample}} - m(\text{CaCO}_3)$  ✓  
 (i) Final answer: 0,25 g ✓ (Range: 0,2 g to 0,3 g)

**Nasienkriteria:**

- (a) Enige formule:  $\text{pH} = -\log[\text{H}_3\text{O}^+]/\text{pH} = -\log[\text{H}^+]/[\text{H}_3\text{O}^+] = 10^{-\text{pH}}$  ✓  
 (b) Vervang 1,61 in  $\text{pH} = -\log[\text{H}_3\text{O}^+]$  ✓  
 (c) Bereken  $n(\text{HCl})_{\text{ongebruik}}$  using  $c = \frac{n}{V}$  ✓  
 (d) Bereken  $n(\text{HCl})_{\text{begin}}$  using  $c = \frac{n}{V}$  ✓  
 (e) Bereken  $n(\text{HCl})_{\text{gebruik}} = n(\text{HCl})_{\text{begin}} - n(\text{HCl})_{\text{ongebruik}}$  ✓  
 (f) Gebruik ratio 1:2 van HCl **GEBRUIK** van (e) om  $n(\text{CaCO}_3)$  te bereken ✓  
 (g) Vervang 100 EN  $n(\text{CaCO}_3)$  van (f) in  $n = \frac{m}{M}$  ✓  
 (h) Massa of onsuiverheid =  $m_{\text{monster}} - m(\text{CaCO}_3)$  ✓  
 (i) Finale antwoord: 0,25 g ✓ (Gebied: 0,2 g tot 0,3 g)

$\text{pH} = -\log[\text{H}_3\text{O}^+] \quad \checkmark \text{ (a)}$   
 $\checkmark \text{ (b)} \quad 1,61 = -\log[\text{H}_3\text{O}^+]$   
 $[\text{H}_3\text{O}^+] = 10^{-1,61}$   
 $= 2,45 \times 10^{-2} \text{ mol} \cdot \text{dm}^{-3} \text{ (0,0245)}$   
 $n(\text{HCl})_{\text{unused}} = n(\text{H}_3\text{O}^+) = cV$   
 $= (2,45 \times 10^{-2})(0,2) \quad \checkmark \text{ (c)}$   
 $= 4,9 \times 10^{-3} \text{ mol (0,0049)}$   
 $n(\text{HCl})_{\text{initial}} = cV$   
 $= (0,15)(0,2) \quad \checkmark \text{ (d)}$   
 $= 3 \times 10^{-2} \text{ mol (0,03)}$   
 $n(\text{HCl})_{\text{used}} = 3 \times 10^{-2} - 4,9 \times 10^{-3} \quad \checkmark \text{ (e)}$   
 $= 2,51 \times 10^{-2} \text{ mol (0,0251)}$   
 Reaction ratio  $n\text{CaCO}_3 : n\text{HCl} = 1:2$   
 $n(\text{CaCO}_3) = \frac{1}{2}(2,51 \times 10^{-2}) \quad \checkmark \text{ (f)}$   
 $= 1,25 \times 10^{-2} \text{ mol}$   
 $n(\text{CaCO}_3) = \frac{m}{M}$   
 $1,25 \times 10^{-2} = \frac{m}{100} \quad \checkmark \text{ (g)}$   
 $m(\text{CaCO}_3) = 1,25 \text{ g}$   
 $m \text{ of impurity in the sample} = 1,5 - 1,25 \quad \checkmark \text{ (h)}$   
 $= 0,25 \text{ g} \quad \checkmark \text{ (i)}$

(9)  
[22]

**QUESTION 8/VRAAG 8**

8.1  $\text{H}^+/\text{H}_3\text{O}^+$  ions/hydrogen ions/hydronium ions/oxonium ions ✓  
*Waterstofione/hidroniumione/oksoniumione* (1)

8.2 0,77 V ✓ (1)

8.3 A ✓ (1)

8.4  $\text{H}_2$  is a stronger reducing agent ✓ than  $\text{Fe}^{2+}/\text{Fe(II)}$  ions ✓ and will reduce  $\text{Fe}^{3+}/\text{Fe(III)}$  ions ✓ (to  $\text{Fe}^{2+}/\text{Fe(II)}$  ions). 

*$\text{H}_2$  is 'n sterker reduseermiddel as  $\text{Fe}^{2+}/\text{Fe(II)}$ -ione en sal  $\text{Fe}^{3+}/\text{Fe(III)}$ -ione reduseer (na  $\text{Fe}^{2+}/\text{Fe(II)}$ -ione).*

**OR/OF**

$\text{Fe}^{2+}$ -ion is a weaker reducing agent ✓ than  $\text{H}_2$  ✓ and therefore  $\text{Fe}^{3+}/\text{Fe(III)}$  ions (to  $\text{Fe}^{2+}/\text{Fe(II)}$  ions) will be reduced. ✓ 

*$\text{Fe}^{2+}$ -ioon is 'n swakker reduseermiddel as  $\text{H}_2$  en sal  $\text{Fe}^{3+}/\text{Fe(III)}$ -ione reduseer (na  $\text{Fe}^{2+}/\text{Fe(II)}$ -ione).* (3)

8.5

8.5.1 Pt/Platinum ✓ (1)

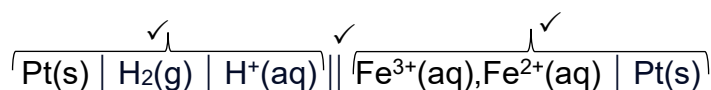
8.5.2  $\text{H}_2 \rightarrow 2\text{H}^+ + 2\text{e}^-$  ✓✓

**NOTE/NOTA:**

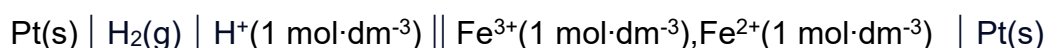
- $2\text{H}^+ + 2\text{e}^- \leftarrow \text{H}_2$  ( $\frac{2}{2}$ )  
 $\text{H}_2 \rightleftharpoons 2\text{H}^+ + 2\text{e}^-$  ( $\frac{1}{2}$ )  
 $2\text{H}^+ + 2\text{e}^- \rightleftharpoons \text{H}_2$  ( $\frac{0}{2}$ )  
 $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$  ( $\frac{0}{2}$ )
- Ignore if charge omitted on electron./Ignoreer indien lading weggelaat op elektron.
- If charge (+) omitted on  $\text{H}^+$ /Indien lading (+) weggelaat op  $\text{H}^+$ :  
 Example/Voorbeeld:  $\text{H}_2 \rightarrow 2\text{H} + 2\text{e}^-$  Max/Maks:  $\frac{1}{2}$

(2)

8.5.3



**OR/OF**



**ACCEPT/AANVAAR:**



(3)

8.6 The reaction reaches equilibrium/no charges/electrons flow. ✓

*Die reaksie bereik ewewig/geen ladings/elektrone vloei.*

(1)

**[13]**

**QUESTION 9/VRAAG 9****9.1 ANY ONE:**

- The (chemical) process in which electrical energy is converted to chemical energy. ✓✓ **(2 or 0)**
- The use of electrical energy to produce a chemical change.
- Decomposition of an ionic compound by means of electrical energy.
- The process during which an electric current passes through a solution/ionic liquid/molten ionic compound.

**ENIGE EEN:**

- Die (chemiese) proses waarin elektriese energie omgeskakel word na chemiese energie. ✓✓ **(2 of 0)**
- Die gebruik van elektriese energie om 'n chemiese verandering teweeg te bring.
- Ontbinding van 'n ioniese verbinding met behulp van elektriese energie.
- Die proses waardeur 'n elektriese stroom deur 'n oplossing/ioniese vloeistof/gesmelte ioniese verbinding beweeg.

(2)

**9.2**  $2\text{H}_2\text{O}(\ell) + 2\text{e}^- \rightarrow \text{H}_2(\text{g}) + 2\text{OH}^-(\text{aq})$  ✓✓**NOTE/NOTA:**

- $\text{H}_2(\text{g}) + 2\text{OH}^-(\text{aq}) \leftarrow 2\text{H}_2\text{O}(\ell) + 2\text{e}^-$  ( $\frac{2}{2}$ )
  - $2\text{H}_2\text{O}(\ell) + 2\text{e}^- \rightleftharpoons \text{H}_2(\text{g}) + 2\text{OH}^-(\text{aq})$  ( $\frac{1}{2}$ )
  - $\text{H}_2(\text{g}) + 2\text{OH}^-(\text{aq}) \rightleftharpoons 2\text{H}_2\text{O}(\ell) + 2\text{e}^-$  ( $\frac{0}{2}$ )
  - $2\text{H}_2\text{O}(\ell) + 2\text{e}^- \leftarrow \text{H}_2(\text{g}) + 2\text{OH}^-(\text{aq})$  ( $\frac{0}{2}$ )
  - Ignore if charge omitted on electron. / Ignoreer indien lading weggelaat op elektron.
  - If charge (-) omitted on  $\text{OH}^-$  / Indien lading (-) weggelaat op  $\text{OH}^-$
- Example/Voorbeeld:  $2\text{H}_2\text{O}(\ell) + 2\text{e}^- \rightarrow \text{H}_2(\text{g}) + 2\text{OH}(\text{aq})$  ✓
- Max./Maks:  $\frac{1}{2}$
- Ignore phases / Ignoreer fases

(2)

**9.3** X to Y / X tot Y ✓

(1)

9.4

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Marking criteria:</b></p> <p>(a) Substitute <math>300 \times 10^{-3}</math> and <math>24 \text{ dm}^3</math> into</p> $n = \frac{V}{V_m} \checkmark$ <p>(b) Using ratio 1:2 to calculate <math>n(e^-)</math> ✓</p> <p>(c) Substitute <math>6,02 \times 10^{23} \text{ mol}^{-1}</math> in</p> $n = \frac{N}{N_A} \checkmark$ <p>(d) Final correct answer:</p> <p><math>1,505 \times 10^{22}</math> electrons ✓</p> <p>Range: <math>1,505 \times 10^{22}</math> to <math>2,41 \times 10^{22}</math> electrons</p> | <p><b>Nasienkriteria:</b></p> <p>(a) Vervang <math>300 \times 10^{-3}</math> en <math>24 \text{ dm}^3</math> in</p> $n = \frac{V}{V_m} \checkmark$ <p>(b) Gebruik verhouding 1:2 om <math>n(e^-)</math> te bereken ✓</p> <p>(c) Vervang <math>6,02 \times 10^{23} \text{ mol}^{-1}</math> in</p> $n = \frac{N}{N_A} \checkmark$ <p>(d) Finale korrekte antwoord:</p> <p><math>1,505 \times 10^{22}</math> elektrone ✓</p> <p>Gebied: <math>1,505 \times 10^{22}</math> tot <math>2,41 \times 10^{22}</math> elektrone</p> |
| <p><b>OPTION 1/OPSIE 1:</b></p> $n(\text{Cl}_2) = \frac{V}{V_m}$ $= \frac{300 \times 10^{-3}}{24} \checkmark (\text{a})$ $= 0,0125 \text{ mol (0,01)}$ <p style="text-align: center;">↓</p> $n(e^-) = 2n(\text{Cl}_2)$ $= 2(0,0125) \checkmark (\text{b})$ $= 0,025 \text{ mol}$ <p style="text-align: center;">↓</p> $n(e^-) = \frac{N}{N_A}$ <p style="text-align: center;">↓</p> $0,025 = \frac{N}{6,02 \times 10^{23}} \checkmark (\text{c})$ $N = 1,505 \times 10^{22} \text{ (electrons)} \checkmark (\text{d})$ | <p><b>OPTION 2/OPSIE 2:</b></p> $n(\text{Cl}_2) = \frac{V}{V_m}$ $= \frac{300 \times 10^{-3}}{24} \checkmark (\text{a})$ $= 0,0125 \text{ mol (0,01)}$ <p style="text-align: center;">↓</p> $n(\text{Cl}_2) = \frac{N}{N_A}$ $0,0125 = \frac{N}{6,02 \times 10^{23}} \checkmark (\text{c})$ $N = 7,525 \times 10^{21} (\text{Cl}_2)$ <p style="text-align: center;">↓</p> $N(e^-) = 2n(\text{Cl}_2)$ $= 2(7,525 \times 10^{21}) \checkmark (\text{b})$ $= 1,505 \times 10^{22} \text{ (electrons)} \checkmark (\text{d})$ |

TOTAL/TOTAAL:

(4)  
[9]  
150